25901857 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION April 2026

second SEMESTER

Skill Development Components (SDC) - MG2SDCFTA102 - FOOD ADDITIVES AND PACKAGING

(2025 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One Sentence

Answer any **10** questions

Each question carries **1** marks

1. List the classification of food additives based on intention. [K] / [CO1]
2. Recall two control measures for preventing unintentional additives. [K] / [CO1]
3. Define E-numbering. [K] / [CO1]
4. Define acidity regulators with an example. [K] / [CO1]
5. State the source of carotene used as a food colour. [K] / [CO2]
6. Define low-calorie sweeteners. [K] / [CO2]
7. Explain briefly how antioxidants prevent oxidation. [U] / [CO2]
8. Name any four humectants used in food products. [K] / [CO2]
9. Define buffer solution. [K] / [CO2]
10. List the properties of TFS used in packaging. [K] / [CO3]
11. List two advantages of Controlled Atmosphere Packaging. [K] / [CO3]

12. Name any two chemical sterilizing agents used in aseptic packaging. [K] / [CO3]

[1x10 = 10]

Part B

Short Answer

Answer any **4** questions

Each question carries **5** marks

13. Define JECFA and CAC. [K] / [CO1]

14. Define chemical leavening agents with suitable examples. [K] / [CO2]

15. Differentiate between natural and synthetic curing agents with suitable examples. [U] / [CO2]

16. Discuss the importance of sequestrants in maintaining quality and shelf life of processed foods. [U] / [CO2]

17. Outline the main functions of each type of packaging system. [U] / [CO3]

18. Describe the method used to measure thickness using a micrometer. [U] / [CO3]

[5x4 = 20]

Part C

Essay

Answer any **2** questions

Each question carries **10** marks

19. List out the properties of food additives. [K] / [CO1]

20. Define thickening agents in detail along with its types and functions in food. [U] / [CO2]

21. Classify preservatives into Class I and Class II along with their functions, examples, and permitted levels. [U] / [CO2]

22. Explain the concept of intelligent packaging and describe the different types of indicators used in it. [U] / [CO3]

[10x2 = 20]